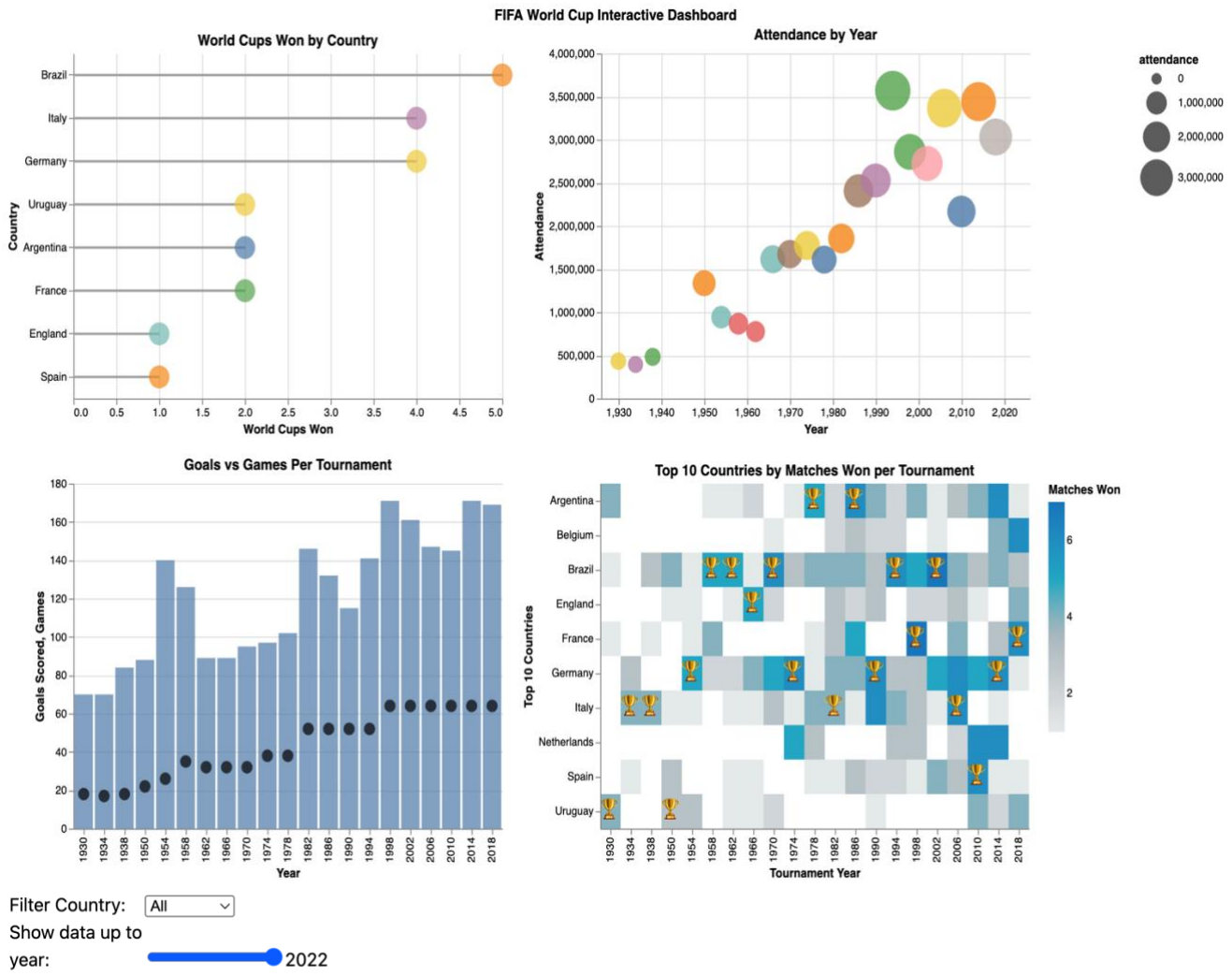
COMP40610 Visual Exploration Tool Design Document

Student Name: Vishal Goyal
Student Number: 252712812

Title:
FIFA World Cup Interactive Dashboard

Screenshot:



Dataset Overview:

This dashboard visualizes FIFA World Cup history using two datasets from the TidyTuesday 2022 collection. The first, worldcups.csv, contains tournament-level details such as host country, total attendance, games played, and total goals. The second, wcmatches.csv, includes match-level results, winning teams, and match specifics. Both sources, available in the project's repository (see: TidyTuesday FIFA-World-Cup 2022 Dataset https://github.com/rfordatascience/tidytuesday/tree/main/data/2022/2022-11-29), were

selected for their comprehensive and varied statistics. Using Vega-Lite, the dashboard links several interactive charts, letting users explore country dominance, attendance, and match outcomes over time. The aim is to make historical World Cup data accessible and engaging for deeper exploration.

Design Consideration:

Overall Goal: This dashboard is designed for interactive exploration of FIFA World Cup history, connecting team achievements, tournament statistics, and long-term patterns. The audience can use it to answer questions like:

- Which countries dominate World Cup history?
- How has goal scoring evolved over time?
- In which years did specific countries perform exceptionally?

Lollipop Chart: World Cups Won by Country

This chart visualises the total number of World Cup tournaments won by each nation. A lollipop chart was selected because it effectively highlights ranking while maintaining a clean visual structure, making it easier to compare countries with close win counts. The length of the rule encodes the quantitative variable *wins*, while the circular marker reinforces the final value, improving readability. I chose this chart instead of a horizontal bar chart because lollipops reduce visual ink and focus attention on the key metric rather than bar thickness. Alternatives considered included a bar chart and a bubble chart; bar charts were more cluttered at this scale, and bubble charts make precision comparisons harder due to varying circle areas. The lollipop design offers an ideal balance of minimalism, clarity, and emphasis on relative differences.

Scatter Plot: Attendance by Year

This scatter plot visualises World Cup attendance over the years, allowing the audience to observe long-term patterns and pinpoint tournaments with unusually high or low attendance. The year is mapped to the X-axis, attendance to the Y-axis, and the size of each circle reflects attendance figures, enhancing visual emphasis on magnitude. Scatter plots were selected because they are well-suited for analysing temporal trends and spotting outliers, particularly given the World Cup's irregular scheduling (every four years, with some interruptions). A line chart was considered but rejected: it suggests continuity between tournaments, which does not accurately reflect the event structure. Bubble charts were also evaluated, but proved too cluttered when representing multiple variables. The scatter format provides the cleanest, most interpretable view.

Bar + Scatter Plot: Goals vs Games Per Tournament

This chart integrates a bar chart, showing total goals scored in each tournament, with a scatter plot representing the number of games played, both aligned along the same X-axis. Its main goal is to illustrate how World Cup formats and scoring rates have changed over time. Bars display the number of goals, while dots indicate total games, making it easy to see whether increases in goals are due to more matches or shifts in playing style. Although a dual-axis chart was considered, it was dismissed because overlapping lines for separate variables can cause confusion and obscure the data. By combining bars and points, the chart keeps the two metrics distinct yet easily comparable. Interactive hover features provide precise values, giving users additional detail as needed.

Heatmap: Top 10 Countries by Matches Won per Tournament

This heatmap visualises the number of matches won per World Cup year by the top ten most successful countries. The X-axis encodes tournament year, while the Y-axis encodes country. Colour intensity represents the number of matches won, enabling users to quickly identify dominant years or periods of strong performance. The heatmap supports categorical × temporal comparisons far better than tables or line charts, which become unreadable with many categories. A significant enhancement is the inclusion of trophy icons marking years in which a country won the World Cup. This provides both quantitative and categorical information in a single unified visual. Alternatives considered included a choropleth map, since a geographic view fits the global nature of the World Cup. However, it was not feasible due to limitations in the dataset: the match data does not include country codes or geographic identifiers. The heatmap offers a dense, information-rich yet readable representation of multi-year tournament performance.

Interaction Consideration:

The dashboard features two main interactive controls: a country filter and a year slider. These tools update every chart at once, enabling users to examine data for a particular country or analyse trends up to a selected year. Implemented with Vega-Lite parameters, these controls directly filter the dataset, ensuring the dashboard remains fast and responsive.

Hover tooltips offer extra information without overcrowding the charts. While more advanced interactions—like brushing or individual chart filters—were considered, they were ultimately excluded to avoid unnecessary complexity and because they did not enhance the user experience. This approach ensures the interactions remain straightforward, intuitive, and consistent throughout the dashboard.